

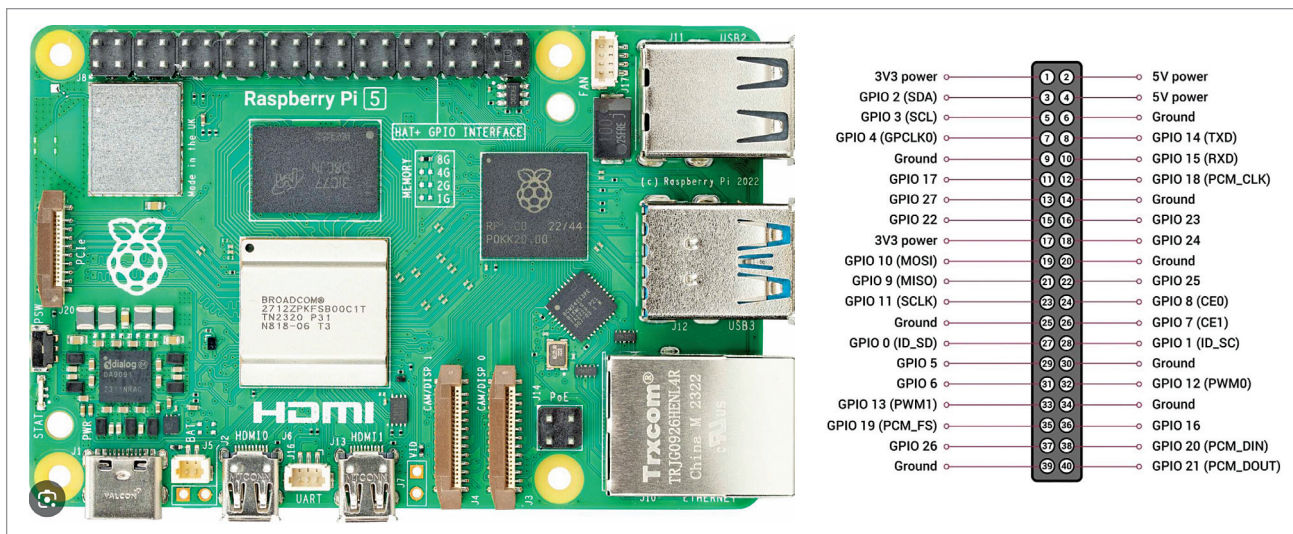


Fig. 3: Adhesive-type metallic heatsink attached to the Hailo-8L AI processor

BILL OF MATERIALS

Components	Quantity
Raspberry Pi-5	1
Hailo-8L AI accelerator	1
Micro gear servo motor	2
Raspberry Pi heat sink and fan	1
5V 3A Raspberry Pi power adaptor	1
HDMI display	1
OLED display	1
Raspberry Pi camera	1
PCA-9685 servo driver	1

improved heat dissipation, attaching a small adhesive-type metallic heatsink directly to the Hailo-8L AI processor is advisable, as shown in Fig. 3.

Pan-tilt mechanism

Pan-tilt servo brackets are available in various designs. For initial setups, a lightweight plastic bracket driven by a pair of MG90S servos is sufficient and cost-effective. For more robust applications, heavy-duty metallic pan-tilt brackets are available, but these require equally strong servos, such as the metallic MG995, to handle the load.

Heavy-duty pan-tilt mechanisms are recommended only when operating with the Raspberry Pi 5 and Hailo-8L platform. This configuration operates at high speed, and when no face is detected, the system

rapidly criss-crosses the frame, as illustrated in Fig. 4.

For servo operation, a PCA9685 board is used to simplify servo control. It communicates over the I²C interface, freeing up GPIO pins and allowing precise, simultaneous control of multiple motors. In this configuration, two channels are used for pan and tilt, with the remaining channels reserved for future expansion, such as robotic arms or wheels.

To use the PCA9685, the I²C interface on the Raspberry Pi must be enabled. This can be done by running `sudo raspi-config`, selecting Interface Options, and enabling I²C.

Display issue

For the default Hailo-8L setup, a display is mandatory. Without a display connected and powered on, Hailo programs do not run. Any small TFT display, such as a 5-inch (12.7cm) or 7-inch (17.7cm) HDMI display, or a larger screen such as a 4K wall-mounted television, can be used. While display output is generally stable on a television, smaller TFT screens may show instability, such as flickering or blanking, particularly during high-load operations such as AI inference.

Typically, TFT and LCD displays consume substantial power, with

the backlight drawing a significant amount of current. Therefore, the display should be powered by a separate power source, not by the Raspberry Pi 5's USB port.

If the issue persists, adjust the HDMI settings by editing the configuration file. All other lines should remain unchanged. Add the required entries at the end of the file, save the changes, and reboot the system for them to take effect.

```
$> sudo nano /boot/firmware/config.txt
# Bera added these lines

[HDMI:0]
hdmi_force_hotplug=1
hdmi_group=2
hdmi_mode=87
hdmi_drive=2
hdmi_timings=1024 1 48 32 80 600 1 3 4 23
0 0 0 60 0 64000000 3
[HDMI:1] # for your TV (optional, can be
removed if unnecessary)
hdmi_group=1 # CEA mode (for TVs)
hdmi_mode=16 # 1080p @ 60Hz (adjust based
on TV)
hdmi_drive=2
hdmi_force_hotplug=1
config_hdmi_boost=5
```

In a later revision, the code was modified to allow Hailo-8L applications to run in true headless mode without requiring a display to be connected. A small I²C OLED display was used to show the number of detected faces and the